Exponential Families

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Many of the distributions in this course, such as the Bernoulli, Poisson and Normal distributions, share one algebraic form for their probability mass function or density. Distributions with that form are called an exponential family, and they share properties that make estimation easier. This page defines the form and its natural parameter, and checks the form for the Poisson distribution on data.

The exponential family form

Definition 0.1 (Exponential family)**.** Let X be a random variable whose distribution depends on one parameter η, and let $\mathbb{f}(x;\eta)$ be its probability mass function or density. The distributions of X form an **exponential family** if there are a nonnegative function h and a function T of x alone, and a function A of η alone, such that

$$\mathbb{f}(x;\eta) = h(x)\exp\{\eta\, T(x) - A(\eta)\} \tag{1}$$

for every value x and every allowed η.

Remark (Source: Exponential family)

Remark 0.1 (Source: Exponential family). Casella and Berger [1, Sec. 3.4], Eq. 3.4.1 write the family as a sum of k terms inside the exponential, each a function of the parameter times a function of x; Dobson and Barnett [2, Sec. 3.2], Eq. 3.3 write it in the form $\exp\{a(y)b(\theta) + c(\theta) + d(y)\}$. Equation 1 is the case of one term. It is Dobson and Barnett's canonical form when $T(x) = x$, and the natural parameter η plays the role of their $b(\theta)$. See the notation choices for how the symbols line up.

Example 0.1 (The Poisson distribution is an exponential family)**.** Let $X \sim \text{Pois}(\lambda)$, the Poisson distribution with rate $\lambda > 0$. Its probability mass function has one factor free of λ, and the factors λ^x and $\exp\{-\lambda\}$. We rewrite it in the form of Equation 1 by writing λ^x as an exponential of a logarithm:

$$
\begin{aligned}
\mathbb{f}(x;\lambda) &= \frac{\lambda^x \exp\{-\lambda\}}{x!} && \text{(Poisson mass function)} \\
&= \frac{1}{x!} \exp\{\log\{\lambda^x\}\} \exp\{-\lambda\} && \text{(a positive number is the exponential of its logarithm)} \\
&= \frac{1}{x!} \exp\{x \log\{\lambda\}\} \exp\{-\lambda\} && \text{(the logarithm of a power is the exponent times the logarithm)} \\
&= \frac{1}{x!} \exp\{x \log\{\lambda\} - \lambda\} && \text{(the product of exponentials is the exponential of the sum)}
\end{aligned}
$$

So the Poisson distributions are an exponential family with $h(x) = 1/x!$, $T(x) = x$, natural parameter $\eta = \log\{\lambda\}$, and $A(\eta) = \lambda = \exp\{\eta\}$.
To check the algebra on data, we use the number of tropical cyclones in northeastern Australia in each of 13 seasons [2, Sec. 1.6.5], with the rate set to the sample mean. We evaluate both sides of Equation 1 at every observed count.

R

```r
cyclones <- c(6, 5, 4, 6, 6, 3, 12, 7, 4, 2, 6, 7, 4)
rate <- mean(cyclones)

base_measure <- function(x) 1 / factorial(x)
natural_parameter <- log(rate)
log_partition <- function(eta) exp(eta)

exponential_form <- base_measure(cyclones) *
  exp(natural_parameter * cyclones - log_partition(natural_parameter))

data.frame(
  count = cyclones,
  poisson = dpois(cyclones, rate),
  exponential_form = exponential_form
)
#>    count    poisson exponential_form
#> 1      6 0.15764459       0.15764459
#> 2      5 0.17078164       0.17078164
#> 3      4 0.15417787       0.15417787
#> 4      6 0.15764459       0.15764459
#> 5      6 0.15764459       0.15764459
#> 6      3 0.11135069       0.11135069
#> 7     12 0.00683927       0.00683927
#> 8      7 0.12472979       0.12472979
#> 9      4 0.15417787       0.15417787
#> 10     2 0.06031496       0.06031496
#> 11     6 0.15764459       0.15764459
#> 12     7 0.12472979       0.12472979
#> 13     4 0.15417787       0.15417787
max(abs(dpois(cyclones, rate) - exponential_form))
#> [1] 1.11022e-16
```

Python

```python
import numpy as np
import pandas as pd
from scipy.special import factorial
from scipy.stats import poisson

cyclones = np.array([6, 5, 4, 6, 6, 3, 12, 7, 4, 2, 6, 7, 4])
rate = cyclones.mean()

natural_parameter = np.log(rate)
exponential_form = (1 / factorial(cyclones)) * np.exp(
    natural_parameter * cyclones - np.exp(natural_parameter)
)

table = pd.DataFrame(
    {
        "count": cyclones,
        "poisson": poisson.pmf(cyclones, rate),
        "exponential_form": exponential_form,
    }
)
print(table)
#>     count   poisson  exponential_form
#> 0       6  0.157645          0.157645
#> 1       5  0.170782          0.170782
#> 2       4  0.154178          0.154178
#> 3       6  0.157645          0.157645
#> 4       6  0.157645          0.157645
#> 5       3  0.111351          0.111351
#> 6      12  0.006839          0.006839
#> 7       7  0.124730          0.124730
#> 8       4  0.154178          0.154178
#> 9       2  0.060315          0.060315
#> 10      6  0.157645          0.157645
#> 11      7  0.124730          0.124730
#> 12      4  0.154178          0.154178
print(np.max(np.abs(poisson.pmf(cyclones, rate) - exponential_form)))
#> 5.551115123125783e-17
```

The two columns agree to within rounding error: the largest difference is 1.1e-16.

Definition 0.2 (Natural parameter)**.** In the form of Equation 1 of an exponential family, the parameter η that multiplies the statistic $T(x)$ inside the exponential is the **natural parameter**, also called the **canonical parameter** [2, Sec. 3.2].

Example 0.2 (The natural parameter of the Poisson distribution is the log rate)**.** In Example 0.1, the Poisson rate λ and the natural parameter η are related by

$$\eta = \log\{\lambda\} \qquad \text{and} \qquad \lambda = \exp\{\eta\}$$

For the cyclone data, the rate set to the sample mean is 5.538 cyclones per season, so the natural parameter is $\eta = 1.712$. The natural parameter can be any real number, while the rate must be greater than zero.

Example 0.3 (The Bernoulli distribution is an exponential family)**.**
Let $X \sim \text{Bernoulli}(\pi)$, the Bernoulli distribution with success probability $\pi \in (0, 1)$. Its probability mass function is $\pi^x(1-\pi)^{1-x}$ for $x \in \{0, 1\}$. We rewrite it in the form of Equation 1 by collecting the factors that have the exponent x:

$$\begin{aligned}
\mathbb{f}(x;\pi) &= \pi^x(1-\pi)^{1-x} && \text{(Bernoulli mass function)} \\
&= \pi^x(1-\pi)^{-x}(1-\pi) && \text{(a power with a sum of exponents is the product of the powe} \\
&= \left(\frac{\pi}{1-\pi}\right)^x(1-\pi) && \text{(powers with the same exponent combine)} \\
&= \exp\left\{\log\left\{\left(\frac{\pi}{1-\pi}\right)^x\right\}\right\}(1-\pi) && \text{(a positive number is the exponential of its logarithm)} \\
&= \exp\left\{x\log\left\{\frac{\pi}{1-\pi}\right\}\right\}(1-\pi) && \text{(the logarithm of a power is the exponent times the logarithı} \\
&= \exp\left\{x\log\left\{\frac{\pi}{1-\pi}\right\}\right\}\exp\{\log\{1-\pi\}\} && \text{(a positive number is the exponential of its logarithm)} \\
&= \exp\left\{x\log\left\{\frac{\pi}{1-\pi}\right\} + \log\{1-\pi\}\right\} && \text{(the product of exponentials is the exponential of the sum)}
\end{aligned}$$

The statistic is $T(x) = x$, and $h(x) = 1$. The natural parameter is the log odds, $\eta = \log\{\pi/(1-\pi)\}$. Inverting it gives $\pi = \exp\{\eta\}/(1+\exp\{\eta\})$, so $1-\pi = 1/(1+\exp\{\eta\})$, and the last term in the exponent is $\log\{1-\pi\} = -\log\{1+\exp\{\eta\}\}$. Hence the Bernoulli distributions are an exponential family with $A(\eta) = \log\{1+\exp\{\eta\}\}$.
To check the algebra on data, we use the `birthwt` data, which record the birth weight of babies born at Baystate Medical Center in Springfield, Massachusetts [3], as distributed in R's `MASS` package [4]. The variable `low` is 1 for a baby with low birth weight and 0 otherwise. We set the success probability to the sample mean of `low`, and evaluate both sides of Equation 1 at the two possible values of x.

R

```r
low <- MASS::birthwt$low
prob <- mean(low)

natural_parameter <- log(prob / (1 - prob))
exp_natural_parameter <- exp(natural_parameter)
log_partition <- log(1 + exp_natural_parameter)

x <- c(0, 1)
exponent <- natural_parameter * x - log_partition
exponential_form <- exp(exponent)
bernoulli <- dbinom(x, size = 1, prob = prob)

abs_difference <- abs(bernoulli - exponential_form)
max_difference <- max(abs_difference)

tibble::tibble(x, bernoulli, exponential_form)
#> # A tibble: 2 x 3
#>       x bernoulli exponential_form
#>   <dbl>     <dbl>            <dbl>
#> 1     0     0.688            0.688
#> 2     1     0.312            0.312
max_difference
#> [1] 1.11022e-16
```

Python

```
import numpy as np
import pandas as pd
from scipy.stats import bernoulli

birthwt = pd.read_csv("data/birthwt.csv")
prob = birthwt["low"].mean()

natural_parameter = np.log(prob / (1 - prob))
log_partition = np.log(1 + np.exp(natural_parameter))

x = np.array([0, 1])
exponential_form = np.exp(natural_parameter * x - log_partition)

table = pd.DataFrame(
    {
        "x": x,
        "bernoulli": bernoulli.pmf(x, prob),
        "exponential_form": exponential_form,
    }
)
print(table)
#>    x  bernoulli  exponential_form
#> 0  0   0.687831          0.687831
#> 1  1   0.312169          0.312169
print(np.max(np.abs(bernoulli.pmf(x, prob) - exponential_form)))
#> 2.220446049250313e-16
```

The value of `prob` is 0.312, so the natural parameter is $\eta = -0.79$. The two columns agree to within rounding error: the largest difference is 1.1e-16.

Example 0.4 (The Gaussian distribution with known variance is an exponential family)**.**

Let $X \sim \mathrm{N}(\mu, \sigma^2)$, the Gaussian distribution with mean $\mu \in \mathbb{R}$ and a variance $\sigma^2 > 0$ that we treat as known. Its density is $(2\pi\sigma^2)^{-1/2}\exp\{-(x-\mu)^2/(2\sigma^2)\}$. We rewrite it in the form of Equation 1 by expanding the square in the exponent and separating the factors that involve μ:

$$
\begin{aligned}
\mathbb{f}(x;\mu) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} && \text{(Gaussian density)} \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{x^2 - 2x\mu + \mu^2}{2\sigma^2}\right\} && \text{(the square of a difference)} \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{x^2}{2\sigma^2} + \frac{x\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2}\right\} && \text{(divide each term by the denominator)} \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{x^2}{2\sigma^2}\right\} \exp\left\{\frac{\mu}{\sigma^2}x - \frac{\mu^2}{2\sigma^2}\right\} && \text{(the exponential of a sum is the product of the exponentia}
\end{aligned}
$$

The statistic is $T(x) = x$, and $h(x) = \exp\{-x^2/(2\sigma^2)\}/\sqrt{2\pi\sigma^2}$, which does not involve μ. The natural parameter is $\eta = \mu/\sigma^2$. Inverting it gives $\mu = \sigma^2\eta$, so $\mu^2/(2\sigma^2) = \sigma^2\eta^2/2$. Hence the Gaussian distributions with a known variance are an exponential family with $A(\eta) = \sigma^2\eta^2/2$.

To check the algebra on data, we use the birth weights in grams from the `birthwt` data [3], as distributed in R's `MASS` package [4]. We set μ to the sample mean and σ^2 to the sample variance of the birth weights, and evaluate both sides of Equation 1 at every observed birth weight.

R

```r
bwt <- MASS::birthwt$bwt
mean_bwt <- mean(bwt)
sd_bwt <- sd(bwt)
variance <- sd_bwt^2

natural_parameter <- mean_bwt / variance
log_partition <- variance * natural_parameter^2 / 2

base_exponent <- -bwt^2 / (2 * variance)
base_scale <- sqrt(2 * pi * variance)
base_measure <- exp(base_exponent) / base_scale

exponent <- natural_parameter * bwt - log_partition
exponential_form <- base_measure * exp(exponent)
normal <- dnorm(bwt, mean = mean_bwt, sd = sd_bwt)

relative_difference <- abs(normal - exponential_form) / normal
max_relative_difference <- max(relative_difference)

tibble::tibble(bwt, normal, exponential_form)
#> # A tibble: 189 x 3
#>      bwt   normal exponential_form
#>    <int>    <dbl>            <dbl>
#>  1  2523 0.000463         0.000463
#>  2  2551 0.000473         0.000473
#>  3  2557 0.000475         0.000475
#>  4  2594 0.000487         0.000487
#>  5  2600 0.000489         0.000489
#>  6  2622 0.000496         0.000496
#>  7  2637 0.000501         0.000501
#>  8  2637 0.000501         0.000501
#>  9  2663 0.000508         0.000508
#> 10  2665 0.000508         0.000508
#> # i 179 more rows
max_relative_difference
#> [1] 3.52031e-15
```

Python

```python
import numpy as np
import pandas as pd
from scipy.stats import norm

birthwt = pd.read_csv("data/birthwt.csv")
bwt = birthwt["bwt"]
mean_bwt = bwt.mean()
variance = bwt.var(ddof=1)

natural_parameter = mean_bwt / variance
log_partition = variance * natural_parameter**2 / 2

base_measure = np.exp(-bwt**2 / (2 * variance)) / np.sqrt(2 * np.pi * variance)
exponential_form = base_measure * np.exp(natural_parameter * bwt - log_partition)
normal = norm.pdf(bwt, loc=mean_bwt, scale=np.sqrt(variance))

table = pd.DataFrame(
    {"bwt": bwt, "normal": normal, "exponential_form": exponential_form}
)
print(table.head())
#>     bwt    normal  exponential_form
#> 0  2523  0.000463          0.000463
#> 1  2551  0.000473          0.000473
#> 2  2557  0.000475          0.000475
#> 3  2594  0.000487          0.000487
#> 4  2600  0.000489          0.000489
print(np.max(np.abs(normal - exponential_form) / normal))
#> 3.324739754359985e-15
```

For these data, $\eta = 0.00554$ per gram. The two columns agree to within rounding error: the largest relative difference is 3.5e-15.

Further reading

- The Fall 2024 course site for UC Berkeley's Stat 210A, *Theoretical Statistics* (Will Fithian) has a course reader on exponential families, sufficiency, completeness and unbiased estimation.

References

[1] G. Casella and R. Berger, *Statistical inference*, 2nd ed. Cengage Learning, 2002. Available: https://www.cengage.com/c/statistical-inference-2e-casella-berger/9780534243128/

[2] A. J. Dobson and A. G. Barnett, *An introduction to generalized linear models*, 4th ed. CRC press, 2018. doi: 10.1201/9781315182780.

[3] D. W. Hosmer and S. Lemeshow, *Applied logistic regression*. New York: Wiley, 1989.

[4] W. N. Venables and B. D. Ripley, *Modern applied statistics with S*, 4th ed. New York: Springer, 2002. doi: 10.1007/978-0-387-21706-2.